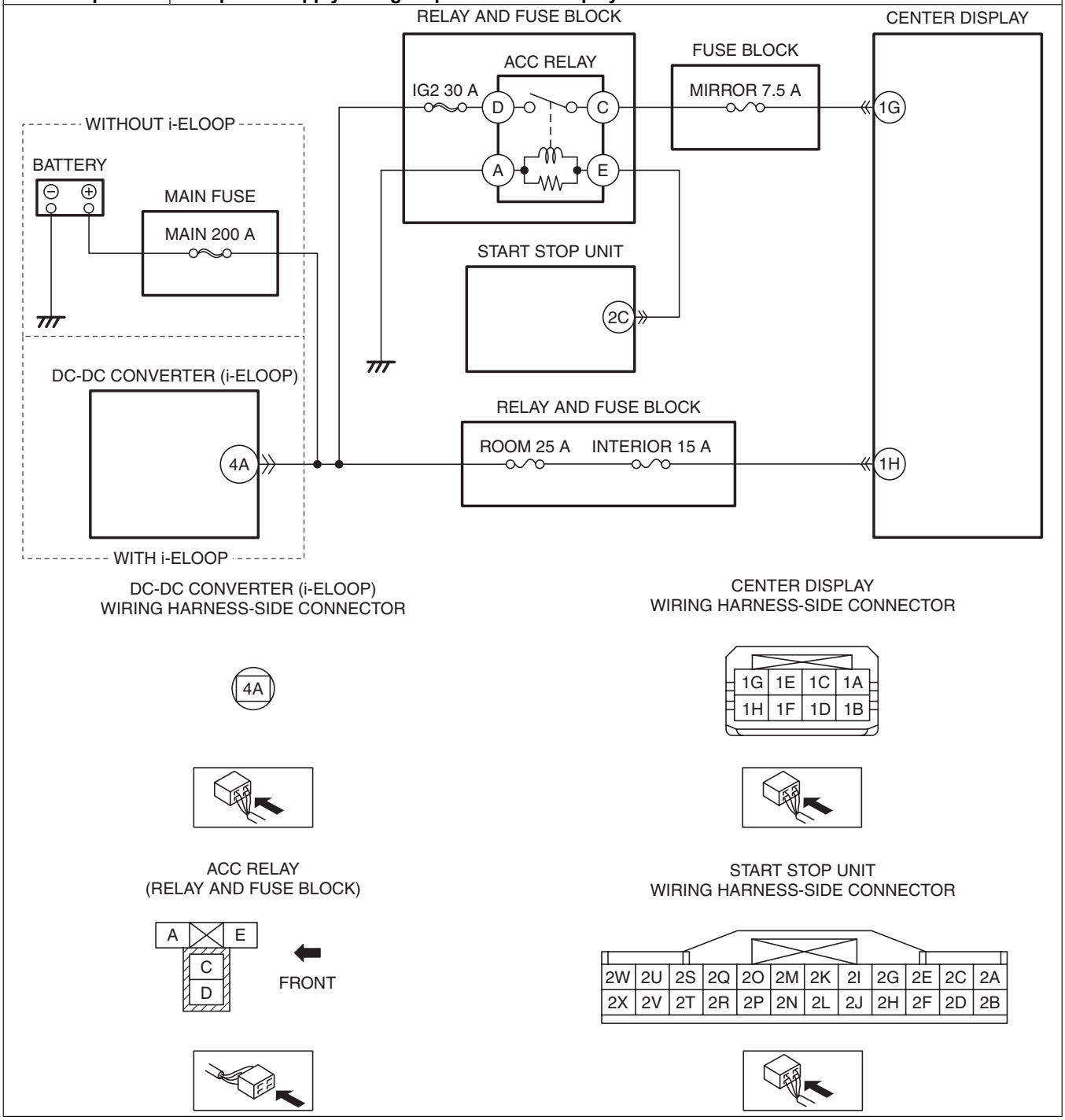
DTC B108E:1C [CONNECTIVITY MASTER UNIT]

id0902n7103400

Description	Low power supply voltage input to center display
Detection condition	• The connectivity master unit (CMU) detected any of the following conditions for **5 s or more** with the ignition switched ON (engine off or on). — TFT LCD driver voltage **3.8 V or less** — LED drive voltage **14.0 V or less** — Touch panel IC voltage **3.9 V or less**
Fail-safe	• Stops operation of the center display.
Possible cause	• DTCs are stored in the PCM. • DTCs are stored in the DC-DC converter (i-ELOOP). — DC-DC converter (i-ELOOP) connector or terminal malfunction — DC-DC converter (i-ELOOP) malfunction • Battery malfunction • Generator malfunction • Center display connector or terminal malfunction • Center display B+ power supply circuit malfunction — Without i-ELOOP: • Short to ground in wiring harness between MAIN 200 A fuse and center display terminal 1H • INTERIOR 15 A fuse malfunction • ROOM 25 A fuse malfunction • MAIN 200 A fuse malfunction • Open circuit in wiring harness between battery positive terminal and center display terminal 1H — With i-ELOOP: • Short to ground in wiring harness between ROOM 25 A fuse and center display terminal 1H • INTERIOR 15 A fuse malfunction • ROOM 25 A fuse malfunction • Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and center display terminal 1H • Center display ACC power supply circuit malfunction — Short to ground in wiring harness between MIRROR 7.5 A fuse and center display terminal 1G — MIRROR 7.5 A fuse malfunction — Open circuit in wiring harness between ACC relay terminal C and center display terminal 1G • Center display malfunction • Connectivity master unit (CMU) malfunction

Description	Low power supply voltage input to center display



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	**Without i-ELOOP:** • Go to Step 3. **With i-ELOOP:** • Go to the next step.

Step	Inspection		Action
2	**VERIFY DC-DC CONVERTER (i-ELOOP) DTCs** • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.
3	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 10. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 10. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
5	**INSPECT CENTER DISPLAY CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the center display connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	**Without i-ELOOP:** • Go to the next step. **With i-ELOOP:** • Go to Step 7.
		No	Repair or replace the connector, then go to Step 10.

Step	Inspection		Action
6	**VERIFY CENTER DISPLAY B+ POWER SUPPLY VOLTAGE** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at center display terminal 1H (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the INTERIOR 15 A fuse, ROOM 25 A fuse, and MAIN 200 A fuse. • If any fuse is blown: — Refer to the wiring diagram and verify if there is a common connector between MAIN 200 A fuse and center display terminal 1H. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. • Replace the fuse. • If any fuse is damaged: — Replace the fuse. • If the fuses are normal: — Refer to the wiring diagram and verify if there is a common connector between positive battery terminal and center display terminal 1H. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.

Step	Inspection		Action
7	**VERIFY CENTER DISPLAY B+ POWER SUPPLY VOLTAGE** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at center display terminal 1H (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Disconnect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the INTERIOR 15 A fuse and ROOM 25 A fuse. • If any fuse is blown: — Refer to the wiring diagram and verify if there is a common connector between ROOM 25 A fuse and center display terminal 1H. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for a short to ground. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If any fuse is damaged: — Replace the fuse. • If the fuses are normal: — Refer to the wiring diagram and verify if there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and center display terminal 1H. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Connect the service plug, then go to Step 10. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)

Step	Inspection		Action
8	**VERIFY CENTER DISPLAY ACC POWER SUPPLY VOLTAGE** • Switch the ignition to ACC. • Measure the voltage at center display terminal 1G (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the MIRROR 7.5 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify if there is a common connector between MIRROR 7.5 A fuse and center display terminal 1G. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify if there is a common connector between ACC relay terminal C and center display terminal 1G. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
9	**VERIFY IF MALFUNCTIONING LOCATION IS CENTER DISPLAY DEPENDING ON REPEATABILITY** • Clear the DTC for the connectivity master unit (CMU) using the M-MDS. (See CLEARING DTC [CONNECTIVITY MASTER UNIT].) • Switch the ignition ON (engine off or on) and wait for **12 s or more**. • Retrieve the connectivity master unit (CMU) DTCs using the M-MDS. (See DTC INSPECTION [CONNECTIVITY MASTER UNIT].) • Is the same DTC displayed?	Yes	Replace the center display, then go to the next step. (See CENTER DISPLAY REMOVAL/INSTALLATION.)
		No	Go to Step 11.
10	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the connectivity master unit (CMU) using the M-MDS. (See CLEARING DTC [CONNECTIVITY MASTER UNIT].) • Switch the ignition ON (engine off or on) and wait for **12 s or more**. • Retrieve the connectivity master unit (CMU) DTCs using the M-MDS. (See DTC INSPECTION [CONNECTIVITY MASTER UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the connectivity master unit (CMU). (See CONNECTIVITY MASTER UNIT (CMU) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.

Step	Inspection		Action
11	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [CONNECTIVITY MASTER UNIT].)
		No	DTC troubleshooting completed.